Supplementary Material

Reproducibility of Pollution–Health Evidence Synthesis Using LLM-Assisted Screening and Extraction

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Contents

1 PubMed Search Strategy

All abstracts were retrieved from PubMed via the NCBI E-utilities API (`esearch` + `efetch`) on February 11, 2026. Two complementary queries were used:

1.1 Broad Query (573 records)

Listing 1: PubMed broad search query

```
("Particulate Matter"[MeSH] OR "PM2.5" OR "fine particulate matter"
 OR "fine particles" OR "ambient particulate matter")
AND ("Hospitalization"[MeSH] OR "hospital admission"
 OR "hospital admissions" OR "emergency department"
 OR "ED visit" OR "ED visits")
AND ("Respiratory Tract Diseases"[MeSH] OR "respiratory"
 OR "asthma" OR "COPD" OR "pneumonia" OR "bronchitis"
 OR "bronchiolitis")
AND ("time series" OR "time-series" OR "case-crossover"
 OR "distributed lag" OR "Poisson regression" OR "GAM"
 OR "generalized additive model")
```

1.2 Design-Filtered Query (222 records)

Listing 2: PubMed design-filtered query

```
("PM2.5" OR "fine particulate matter"
 OR "particulate matter 2.5")
AND ("respiratory" OR "asthma" OR "COPD" OR "pneumonia")
AND ("hospitalization" OR "hospital admission"
 OR "emergency department")
AND ("time series" OR "time-series" OR "case-crossover")
```

1.3 Exclusion Candidates (475 records)

A third query targeted articles clearly outside scope (animal studies, in-vitro, occupational exposure) to populate the "clearly exclude" category.

1.4 Corpus Assembly

From the merged candidate pool (1,021 unique records after deduplication):

- 100 abstracts meeting all six inclusion criteria were labeled "clearly include"
- 100 abstracts with unambiguous exclusion reasons were labeled "clearly exclude"
- 300 abstracts with borderline characteristics (e.g., mixed pollutants, ambiguous outcomes) were labeled "ambiguous"

The 200 "clear" abstracts were labeled by an automated classification rule based on inclusion-criteria keyword matching, and the 300 "ambiguous" abstracts by the same rule operating in low-confidence regions (see Labeling Guide, Section 4). No abstract in the 500-item corpus received a human label at this stage; human dual-independent labeling was conducted later, on a stratified 100-abstract subset, and is reported in Section 4.

2 Prompt Templates

The exact prompts used for both pipeline stages are reproduced below. These prompts were held constant across all 6 models and all 120 experimental runs.

2.1 Stage A: Abstract Screening Prompt

Listing 3: Screening prompt template (`configs/prompts/screening.txt`)

```
You are an expert epidemiologist conducting a systematic review on
the association between PM2.5 exposure and respiratory
hospitalizations.

Given the following title and abstract of a scientific publication,
determine whether it should be INCLUDED or EXCLUDED from the review
based on the criteria below.

## Inclusion Criteria
1. Original epidemiological study (not a review, editorial, or
   commentary)
2. Exposure: PM2.5 or fine particulate matter (<=2.5 um)
3. Outcome: respiratory hospitalization or emergency department visit
4. Study design: time-series or case-crossover
5. Reports a relative risk (RR), odds ratio (OR), or equivalent
   effect measure with confidence interval
6. Published in English

## Exclusion Criteria
- Reviews, meta-analyses, editorials, or commentaries
- Exposure other than PM2.5 (unless PM2.5 is one of multiple
  exposures analyzed)
- Outcome other than respiratory hospitalization/ED visit
- No quantitative effect estimate reported
- Animal or in-vitro studies

## Input
Title: {title}
Abstract: {abstract}

## Instructions
Respond with a JSON object only. Do not include any text outside
the JSON.

{
  "decision": "include" | "exclude" | "uncertain",
  "confidence": 0.0-1.0,
  "rationale": "brief explanation of decision",
  "exposure": "PM2.5" | "other" | "not_specified",
  "outcome": "respiratory_hospitalization" | "respiratory_ED"
             | "other" | "not_specified",
  "study_design": "time_series" | "case_crossover" | "cohort"
                  | "other" | "not_specified",
  "has_effect_estimate": true | false
}
```

2.2 Stage B: Data Extraction Prompt

Listing 4: Extraction prompt template (`configs/prompts/extraction.txt`)

```
You are an expert epidemiologist extracting quantitative data from a
scientific publication on PM2.5 and respiratory hospitalizations for
a systematic review and meta-analysis.

Given the following title and abstract, extract the structured data
below. If a field is not reported in the abstract, use null.

## Input
Title: {title}
Abstract: {abstract}

## Instructions
Extract ALL reported effect estimates. If multiple estimates are
reported (e.g., different lags, subgroups), extract each as a
separate entry in the "estimates" array.

Respond with a JSON object only. Do not include any text outside
the JSON.

{
  "study_id": "first_author_year",
  "study_location": "city, country",
  "study_period": "YYYY-YYYY",
  "study_design": "time_series" | "case_crossover" | "other",
  "population": "general" | "elderly" | "children" | "other",
  "sample_size": null | integer,
  "estimates": [
    {
      "effect_measure": "RR" | "OR" | "HR" | "IRR" | "other",
      "effect_estimate": float,
      "ci_lower": float,
      "ci_upper": float,
      "ci_level": 95,
      "exposure_increment": "per 10 ug/m3" | "per IQR" | "other",
      "lag": "lag0" | "lag1" | "lag0-1" | "other",
      "outcome_specific": "all_respiratory" | "pneumonia"
                         | "COPD" | "asthma" | "other",
      "covariates": ["temperature", "humidity", "trend",
                     "seasonality", "..."]
    }
  ]
}
```

3 JSON Schemas for Output Validation

All LLM outputs were validated against JSON Schema (Draft-07) definitions. Outputs failing validation were flagged but retained to ensure complete provenance.

3.1 Screening Output Schema

Seven required fields, no additional properties permitted: `decision` (enum: include/exclude/uncertain), `confidence` (0.0–1.0), `rationale` (string, min 10 chars), `exposure`, `outcome`, `study_design` (categorical enums), and `has_effect_estimate` (boolean).

3.2 Extraction Output Schema

Required fields: `study_id` (string), `estimates` (array). Optional metadata: `study_location`, `study_period`, `study_design`, `population`, `sample_size`. Each estimate requires: `effect_measure`, `effect_estimate`, `ci_lower`, `ci_upper` (all with range constraints 0.1–10.0).

Full schemas are available in the repository at `configs/schemas/`.

4 Gold Standard Labeling Protocol

4.1 Screening Labels

The 500-item corpus was labeled by an automated classification rule: the 200 "clear" abstracts (100 include, 100 exclude) by inclusion-criteria keyword matching, and the 300 "ambiguous" abstracts by the same rule in low-confidence regions. The rule made no errors on a held-out 50-abstract validation set, which by the rule of three bounds its true error rate at approximately 6% with 95% confidence; we do not describe it as 100% precise. These labels serve as reference values for accuracy computation but do not affect the primary metric (EMR), which measures self-consistency regardless of ground truth.

4.2 Dual-Independent Human Validation

In response to peer review we validated the automated labels against human judgement on a stratified subset of 100 abstracts (25 clear-include, 25 clear-exclude, 50 ambiguous; stratified by category $\times$ year-tertile, seed=42).

Pre-registration. The protocol, the screening templates, and the agreement script were frozen as an OSF registration on 2026-05-12 (`https://doi.org/10.17605/OSF.IO/FGN3E`). The two label sets were returned on 2026-07-15 and 2026-07-29, so the registration pre-dates all label collection. The registration pre-specified the target (Cohen's $\kappa \geq 0.80$, Cochrane Handbook §4.6.6), the discordance-rate expectation ($< 15\%$), and a contingency for missing the target.

Both pre-specified gates were missed, and we report them rather than drop them: $\kappa = 0.529$ against a target of 0.80, and a discordance rate of 25% against an expectation of $< 15\%$.

Two competing explanations for a low coefficient can be ruled out. The prevalence index is low (0.189), and on the binary collapse PABAK (0.558) is almost identical to κ (0.556), so the value is not depressed by skewed marginals—the "kappa paradox" is not operating. On the three-class scale PABAK is 0.625 against κ=0.529; the gap reflects the chance-agreement term that PABAK discards by construction, not marginal skew, and PABAK is generalised here as $(k\,p_o - 1)/(k-1)$ rather than the k=2 form $2p_o - 1$. Weighted variants shift the estimate by less than 0.02, so the result is not an artefact of the three-class scale. With $n = 100$ the interval spans "fair" through "substantial", so we report the value relative to the threshold and do not assign it a descriptive band.

The disagreement is directional, not diffuse. The binary discordant cells are 2 versus 19 (exact McNemar $p = 2.2 \times 10^{-4}$), and marginal homogeneity is rejected on the 3-class table (Stuart–Maxwell $\chi^2(2) = 16.4$, $p = 2.7 \times 10^{-4}$; Bowker $\chi^2(3) = 17.8$, $p = 4.9 \times 10^{-4}$). Rater 2

Table 1: Stage-A screening agreement between the two human raters ($n = 100$). Analytic intervals use the Fleiss–Cohen–Everitt delta method; the bootstrap uses 10,000 resamples with seed 42.

Quantity	**Value**	**95% CI**
Cohen's κ, 3-class	0.529 (SE 0.074)	[0.383, 0.674]
bootstrap, percentile	0.529	[0.383, 0.670]
bootstrap, BCa	0.529	[0.382, 0.669]
Cohen's κ, binary (include vs. exclude, $n = 95$)	0.556	[0.400, 0.712]
Weighted κ, linear (ordinal include–uncertain–exclude)	0.541	—
Weighted κ, quadratic	0.548	—
Raw agreement	75.0%	—
PABAK, 3-class / binary	0.625 / 0.558	—
Prevalence index / bias index	0.189 / 0.179	—
Test against Cochrane target	$z = -3.65$	$p = 1.3 \times 10^{-4}$

Table 2: Confusion matrix, rater 1 (rows) × rater 2 (columns).

	INCLUDE	**EXCLUDE**	**UNCERTAIN**
INCLUDE	28	2	0
EXCLUDE	19	46	2
UNCERTAIN	2	0	1

returned 49 INCLUDE decisions against rater 1's 30. Systematic directional divergence of this kind indicates that the two raters applied different readings of the written protocol, which is a defect of the instrument rather than noise in the raters.

Of the 25 discordances, 19 run in that same direction, and 17 of those 19 turn on inclusion criterion 5 (quantitative effect estimate). Protocol v1.1 stated the criterion as "reports RR, OR, HR (or equivalent) with 95% CI" in its inclusion list while its exclusion list read "no quantitative effect estimate extractable from the abstract" — the first wording admits a stated intention to report an effect, the second demands the numbers themselves. Compounding this, the v1.1 decision table specified outcomes for failure in two or more criteria and for a borderline single criterion, but not for clear failure in exactly one criterion, which is the configuration of those 17 abstracts.

Table 3: Stratum-specific agreement. In the clear-exclude stratum expected agreement equals one, so κ is undefined despite unanimity; PABAK is reported for that reason.

Stratum	n	**Raw agreement**	**Cohen's κ**	**PABAK**
clear-exclude	25	1.000	undefined	1.000
clear-include	25	0.680	0.359	0.520
ambiguous	50	0.660	0.398	0.490

The gold standard is asymmetrically valid, and this qualifies every accuracy figure in the main text. The exclude side is unanimous: both raters endorsed exclusion for all 25 abstracts the automated rule called clearly excludable. The include side is not: of the 25 abstracts the rule called clearly includable, rater 1 endorsed inclusion for 13 and rater 2 for 21, and 8 of the 25 discordances fall inside that stratum. Reported specificity therefore rests on labels the humans confirm, while reported sensitivity rests on include-side labels they substantially contest and should be read as a lower bound.

Protocol amendment (v1.2). Following item (c) of the registered contingency, criterion 5 was split into three levels. Level 5a: a numeric point estimate with a numeric interval is present, and the criterion is satisfied. Level 5b: the abstract states that the effect was estimated but omits the values, which resolves to *uncertain*, on the reasoning that abstract-only screening cannot verify what the abstract omits and that an EXCLUDE would assert something about the study the available evidence does not support. Level 5c: no estimate and no mention of one, which resolves to *exclude*, because total absence is reasonable evidence that the study produced no extractable estimate. The distinction between 5b and 5c is not which criterion failed but how much the abstract reveals about the failure.

The decision table now also separates structural criteria (original study, $PM_{2.5}$ exposure, respiratory hospitalization, English), where a single clear failure excludes, from conditional criteria (design, effect reporting), where a single failure yields *uncertain* — with level 5c as the stated exception. Precedence when several rows apply is exclude > uncertain > include. The amended table is exhaustive by design: the v1.1 defect was an uncovered case, and raters are now instructed to record *uncertain* and escalate rather than improvise if they meet one.

Recalibration round and its limits. A blinded round restricted to the 25 discordant items is underway; the 75 concordant items are not reopened. This round cannot produce a second estimate of κ, and we do not report one. Conditioning re-measurement on prior disagreement moves a recomputed whole-corpus coefficient upward mechanically—at a sufficiently high resolution rate it would cross the Cochrane threshold by construction, carrying no evidential content. The outcome is therefore reported as post-hoc reconciliation agreement conditional on the initially discordant items, and $\kappa = 0.529$ stands as the study's agreement estimate. Demonstrating that the v1.2 amendment repairs the ambiguity would require a fresh independent sample, which we identify as the appropriate next step rather than claim here.

Discordance resolution. Items still discordant after recalibration go to a consensus meeting; items without consensus are settled by the senior author (Y.d.S.T.) as tie-breaker, as pre-registered. The role is deliberately held by a co-author who did not develop the pipeline being scored against this gold standard, which keeps final adjudication independent of the system under evaluation. Tie-breaks are restricted to residual discordant items, adjudicated against the written criteria with no access to model outputs, and logged with the criterion invoked, so each adjudicated item is auditable. The resulting gold standard records, per abstract, whether its label came from first-round agreement, post-recalibration agreement, consensus, or tie-break.

4.3 Extraction Labels

For the 100 "clearly include" articles, extraction templates were created following standardized rules:

- Prefer the primary/main estimate if multiple are reported
- Prefer all-respiratory over specific conditions (asthma, COPD)
- Prefer single-pollutant model over multi-pollutant
- Prefer lag 0–1 if multiple lags reported without a "best" designation
- Record per 10 μg/m^3 – convert if different increment used
- If abstract reports % change, convert: RR = 1 + (%change/100)

Discordance threshold: > 0.005 for effect estimates, > 0.01 for CI bounds.

5 Example Outputs: Exact Match vs. Near-Miss

To illustrate the nature of extraction non-determinism, we present two representative cases.

5.1 Example 1: Exact Match (Local Model – Gemma 2 9B)

All 10 runs produced identical output (EMR = 1.000):

Listing 5: Gemma 2 9B extraction output (identical across all 10 runs)


```
{
  "study_id": "chen_2017",
  "study_location": "Shanghai, China",
  "study_period": "2013-2015",
  "study_design": "time_series",
  "population": "general",
  "sample_size": null,
  "estimates": [
    {
      "effect_measure": "RR",
      "effect_estimate": 1.023,
      "ci_lower": 1.005,
      "ci_upper": 1.042,
      "ci_level": 95,
      "exposure_increment": "per 10 ug/m3",
      "lag": "lag0-1",
      "outcome_specific": "all_respiratory",
      "covariates": ["temperature", "humidity", "day_of_week",
                     "long_term_trend", "seasonality"]
    }
  ]
}
```


5.2 Example 2: Near-Miss (Cloud API – Claude Sonnet 4.5)

Runs 1–8 and 10 produced one output; Run 9 differed in the `estimates` array:

Listing 6: Claude Sonnet 4.5 – Run 9 extracted 2 estimates instead of 1


```
{
  "study_id": "chen_2017",
  "study_location": "Shanghai, China",
  "study_period": "2013-2015",
  "study_design": "time_series",
  "population": "general",
  "sample_size": null,
  "estimates": [
    {
      "effect_measure": "RR",
      "effect_estimate": 1.023,
      "ci_lower": 1.005,
      "ci_upper": 1.042,
      "ci_level": 95,
      "exposure_increment": "per 10 ug/m3",
      "lag": "lag0-1",
```

```
      "outcome_specific": "all_respiratory",
      "covariates": ["temperature", "humidity", "day_of_week",
                     "long_term_trend", "seasonality"]
    },
    {
      "effect_measure": "RR",
      "effect_estimate": 1.031,
      "ci_lower": 1.008,
      "ci_upper": 1.054,
      "ci_level": 95,
      "exposure_increment": "per 10 ug/m3",
      "lag": "lag2",
      "outcome_specific": "pneumonia",
      "covariates": ["temperature", "humidity", "day_of_week"]
    }
  ]
}
```

Key observation: The metadata fields (`study_id`, `study_location`, `study_design`) are identical across all 10 runs (consistent with BERTScore $F_1 = 0.997$). The variation lies in the `estimates` array—Run 9 extracted an additional sub-group estimate (pneumonia at lag 2) that the other 9 runs did not. This single additional estimate causes the hash-based EMR to register a non-match, while the BERTScore over metadata text remains near-perfect. This exemplifies the "semantic paradox" reported in Section 4.5 of the main paper.

6 Model Configurations

Table 4: Complete model configurations for all six LLMs.

Model	Provider	Temp.	Seed	Max Tokens	Timeout	Version
LLaMA 3 8B	Ollama	0.0	42	4096	600s	`llama3:8b`
Mistral 7B	Ollama	0.0	42	4096	600s	`mistral:7b`
Gemma 2 9B	Ollama	0.0	42	4096	600s	`gemma2:9b`
Claude Sonnet 4.5	Anthropic	0.0	—	4096	90s	`claude-sonnet-4-5-20250929`
Gemini 2.5 Pro	Google	0.0	42	8192†	90s	`gemini-2.5-pro`
GPT-4.1	OpenAI	0.0	42	4096	90s	`gpt-4.1`

†Gemini requires higher `maxOutputTokens` because thinking tokens consume the output budget.

6.1 Provider API Documentation Links

For full reproducibility and to disambiguate the exact API surfaces used in this study, Table 5 lists the official documentation pages for each provider and the local serving runtime, with the access date for each link. This is provided so that future reviewers and replicators can verify (a) exactly which parameters were exposed on each API surface at the time of the study, (b) which parameters were renamed, deprecated, or added subsequently, and (c) whether default values (e.g., for unset `top_p`, `stop_sequences`, or system prompts) have changed silently between the study window and any future replication attempt.

Table 5: Official documentation references for the six deployment stacks evaluated and the desconfound cloud route. URLs and access dates are recorded so that replication studies can pin to the documented behaviour rather than to current docs that may drift.

Provider	Stack / Endpoint	Documentation URL	Access date
Anthropic	Claude Sonnet 4.5 / Messages API v1	`https://docs.anthropic.com/en/api/messages`	2026-02-11
Google	Gemini 2.5 Pro / generateContent (v1)	`https://ai.google.dev/api/rest`	2026-02-11
OpenAI	GPT-4.1 / Chat Completions v1	`https://platform.openai.com/docs/api-reference/chat`	2026-02-11
Ollama (local)	LLaMA 3 8B / Mistral 7B / Gemma 2 9B (v0.15.5)	`https://github.com/ollama/ollama/blob/main/docs/api.md`	2026-02-11
OpenRouter → DeepInfra	`meta-llama/llama-3-8b-instruct` (desconfound route)	`https://openrouter.ai/docs`	2026-04-23

Access dates correspond to the start of data collection for each stack. The full request payload schemas used in this study, including provider-specific defaults explicitly set or accepted, are reproduced as JSON skeletons in our public repository (`configs/request_payloads/`).

7 Provenance Protocol Details

7.1 Call-Level Hashing

Each LLM call produces a **call hash** computed as:

Listing 7: Call hash computation (`src/provenance/hasher.py`)

```python
import hashlib, json

def compute_call_hash(prompt, input_text, model_id, temperature,
                      seed=None):
    fields = {
        "prompt": prompt,
        "input_text": input_text,
        "model_id": model_id,
        "temperature": temperature,
        "seed": seed,
    }
    canonical = json.dumps(fields, sort_keys=True,
                           ensure_ascii=True)
    return hashlib.sha256(canonical.encode()).hexdigest()
```

The use of `sort_keys=True` and `ensure_ascii=True` ensures deterministic serialization across platforms and Python versions.

7.2 Output Hashing

The raw LLM response text is hashed directly:

```python
def compute_output_hash(output_text):
    return hashlib.sha256(output_text.encode()).hexdigest()
```

7.3 Run Cards

Each run produces a structured JSON run card containing:

- Run metadata: model ID, provider, stage, run number
- Configuration: temperature, seed, max tokens
- Execution: timestamps, call counts (total/success/fail), duration
- Provenance: aggregate output hash (SHA-256 of sorted individual hashes joined by pipe delimiter)

7.4 Verification Protocol

If two runs of the same model produce different aggregate hashes, the per-call output hashes are compared to identify exactly which abstracts changed. This enables targeted analysis of non-determinism at the individual-abstract level.

8 Corpus Statistics

Table 6: Corpus descriptive statistics.

Statistic	Value
Total abstracts	500
Unique journals	148
Publication range	1994–2026
Mean abstract length (characters)	1,873
Median abstract length (characters)	1,812
Min abstract length (characters)	423
Max abstract length (characters)	4,291
Clear include	100 (20%)
Clear exclude	100 (20%)
Ambiguous	300 (60%)

9 Experimental Timeline

All 120 experimental runs were conducted within a 3-week window to minimize the risk of silent API model updates:

Table 7: Experiment execution timeline.

Model	Screening	Extraction	Duration
LLaMA 3 8B	Feb 11–12	Feb 12–13	∼91h
Mistral 7B	Feb 13–15	Feb 15–16	∼77h
Gemma 2 9B	Feb 16–17	Feb 17	∼46h
Claude Sonnet 4.5	Feb 18	Feb 18–19	∼12h
Gemini 2.5 Pro	Feb 20	Feb 20–21	∼8h
GPT-4.1	Feb 22	Feb 22–Mar 2	∼6h

10 Software and Dependencies

- **Python**: 3.14.3
- **OS**: macOS Darwin 24.6.0 (Apple M4, 24 GB RAM)
- **Ollama**: v0.15.5 (local model serving)
- **Key dependencies**: numpy $\geq$ 1.26, pandas $\geq$ 2.1, scipy $\geq$ 1.12, jsonschema $\geq$ 4.20, bert-score $\geq$ 0.3.13, transformers $\geq$ 4.36
- **No external SDKs**: All API calls use Python's `urllib.request` to ensure uniform request handling
- **Test suite**: 45 tests (pytest $\geq$ 8.0), 100% pass rate
- **License**: MIT
- **Repository**: `https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility`

11 Complete Bootstrap Confidence Intervals

All confidence intervals were computed using the percentile bootstrap method with 10,000 resamples. For models with EMR = 1.000, the bootstrap trivially returns [1.000, 1.000]; the rule-of-three upper bound on the non-match rate is $3/n$ (0.6% for $n = 500$ screening, 3.0% for $n = 100$ extraction).

12 Fixed-Slot Extraction Prompt and Comparison

The fixed-slot prompt template used in the sensitivity analysis (main text §4.7):

Listing 8: Fixed-slot extraction prompt (`configs/prompts/extraction_fixed_slot.txt`)

```
You are an expert epidemiologist extracting quantitative data from a
scientific publication on PM2.5 and respiratory hospitalizations for
a systematic review and meta-analysis.

Given the following title and abstract, extract the SINGLE PRIMARY effect
estimate using the fixed selection rules below. Do not produce a
variable-length array; produce exactly one primary_estimate object (or
null if no estimate is reportable).

```

Table 8: Complete EMR with 95% bootstrap confidence intervals for all model-stage combinations.

Model	**Stage**	**EMR**	**95% CI**	n
LLaMA 3 8B	Screening	1.000	[1.000, 1.000]*	480
LLaMA 3 8B	Extraction	1.000	[1.000, 1.000]*	96
Mistral 7B	Screening	1.000	[1.000, 1.000]*	500
Mistral 7B	Extraction	1.000	[1.000, 1.000]*	100
Gemma 2 9B	Screening	1.000	[1.000, 1.000]*	500
Gemma 2 9B	Extraction	1.000	[1.000, 1.000]*	100
Claude Sonnet 4.5	Screening	0.974	[0.960, 0.986]	500
Claude Sonnet 4.5	Extraction	0.050	[0.010, 0.090]	100
Gemini 2.5 Pro	Screening	0.936	[0.914, 0.956]	500
Gemini 2.5 Pro	Extraction	0.200	[0.120, 0.280]	100
GPT-4.1	Screening	0.970	[0.954, 0.984]	500
GPT-4.1	Extraction	0.150	[0.080, 0.220]	100

*Rule-of-three upper bound on non-match rate: $3/n$ (screening: 0.6%, extraction: 3.0%).

```
## Fixed selection rules (apply in order)
1. Prefer all-respiratory outcomes over specific subgroups.
2. Prefer single-pollutant PM2.5 model over multi-pollutant adjustments.
3. If multiple lags are reported, prefer lag 0-1 (cumulative); else lag 0.
4. Prefer per 10 ug/m^3 exposure increment; convert if reported per IQR.
5. If percent change reported, convert: RR = 1 + (%change/100).
6. If no quantitative effect estimate with CI is reportable, set
   primary_estimate = null.

## Input
Title: {title}
Abstract: {abstract}

## Output (JSON only, no extra text)
{
  "study_id": "first_author_year",
  "study_location": "city, country",
  "study_period": "YYYY-YYYY",
  "study_design": "time_series" | "case_crossover" | "other",
  "population": "general" | "elderly" | "children" | "other",
  "sample_size": null | integer,
  "primary_estimate": { ... } | null,
  "n_estimates_in_abstract": integer
}
```

13 LLaMA Cloud Desconfound Experiment Details

The desconfound experiment (main text §4.8) serves the identical model weights via cloud and compares against the local Ollama deployment.

Table 9: Fixed-slot extraction sensitivity analysis: 3 cloud models × 3 runs × 100 INCLUDE articles. Variable-length baseline from main Table 3 (10 runs). Provider pinned per call: Claude Sonnet 4.5 via Anthropic API; Gemini 2.5 Pro via Google API; GPT-4.1 via OpenAI API.

Model	**Var-len EMR**	**Fixed-slot EMR**	**Δ abs**	**Δ rel**	**Pairwise dis.**
Claude Sonnet 4.5	0.050	0.030	−0.020	−40.0%	0.78
Gemini 2.5 Pro	0.200	0.185	−0.015	−7.6%	0.49
GPT-4.1	0.150	**0.444**	+0.294	+196.3%	0.36

Table 10: Desconfound experimental configuration. Both deployments use the SAME model weights, prompt, seed=42, and temperature=0; only the serving infrastructure differs.

Parameter	**Local (Ollama)**	**Cloud (DeepInfra)**
Model	`llama3:8b`	`meta-llama/llama-3-8b-instruct`
Provider	Ollama v0.15.5	OpenRouter → DeepInfra (pinned)
Hardware	Apple M4, 24 GB	DeepInfra GPU cluster
Inference	Single-stream	Batched continuous-batching
Seed	42	42
Temperature	0.0	0.0
Max tokens	4,096	512 (screening)
N runs	10	10
N abstracts	500	500
Date range	Feb 11–13, 2026	Apr 25–28, 2026

Table 11: Per-run successful-call counts. Cloud runs 5 and 6 experienced transient infrastructure failures resulting in reduced valid-item counts, which we report transparently as a cloud-reliability finding. Local runs are uniformly 480/500 successful (the same 20 abstracts deterministically exceed response length in every local run).

Run	**Local valid**	**Cloud valid**
1	480	491
2	480	489
3	480	491
4	480	489
5	480	**45***
6	480	**221***
7	480	491
8	480	491
9	480	491
10	480	491

*Transient cloud-side failure (`503 Service Unavailable` bursts).

14 Silver Standards and Cross-Validation

Two independent silver standards complement the heuristic gold standard for the 100 INCLUDE articles.

14.1 Silver-Internal: Majority Vote of Six Evaluated Models

Consensus extraction is computed as the per-field mode across all 6 models × 10 runs (60 votes per item). Numeric fields are binned at 0.01 resolution before mode computation.

Table 12: Silver-internal consensus. Items with consensus = items where at least one model-run produced a non-null value. Mean mode-agreement = average proportion of votes matching the modal value across items with consensus.

Field	Items w/ consensus	Mean mode-agreement
effect_estimate	95/100	0.858
ci_lower	84/100	0.846
ci_upper	84/100	0.845
effect_measure	100/100	0.783
outcome_specific	100/100	0.694
exposure_increment	100/100	0.761
lag	100/100	0.644

14.2 Silver-External: DeepSeek-R1 (5 Runs)

DeepSeek-R1 (`deepseek-reasoner`) is a reasoning model from a different provider family than the 6 evaluated models, providing an independent reference. Five runs were aggregated by per-field majority vote.

Table 13: Silver-external consensus from DeepSeek-R1 (5 runs × 100 items).

Field	Items w/ consensus	Mean mode-agreement
effect_estimate	84/100	0.890
ci_lower	74/100	0.894
ci_upper	74/100	0.889
effect_measure	84/100	0.906
outcome_specific	84/100	0.803
exposure_increment	84/100	0.881
lag	75/100	0.862

For items lacking DeepSeek-R1 consensus across its 5 runs (16 of 100 items on `effect_estimate`; 26 of 100 items on the CI bounds `ci_lower`/`ci_upper`; 25 of 100 on `lag`), per-run accuracy against silver-external is undefined and these items are excluded; the accuracies reported in the corresponding silver-external comparison tables (and the cross-validation table below) are computed over the consensus subset only.

Table 14: Field-level agreement between silver-internal and silver-external (numeric tolerance 0.05; categorical exact match). Spearman ρ on `effect_estimate` = **0.845**.

Field	n compared	Agreement
effect_estimate	84	0.798
ci_lower	74	0.797
ci_upper	74	0.784
effect_measure	84	0.738
outcome_specific	84	0.655
exposure_increment	84	0.798
lag	75	0.653

14.3 Cross-Validation Between Silvers

15 Pairwise Disagreement and Rationale Similarity

Table 15: Pairwise disagreement (mean fraction of $\binom{10}{2}$=45 run-pairs producing different outputs per item; 0 = perfect determinism). Compare to EMR (Table 10): EMR collapses items into binary match/non-match while pairwise disagreement reports the dispersion within non-matches.

Model	Stage	EMR	Pairwise	% items any-disagree
LLaMA 3 8B	screening	1.000	0.0000	0.00%
LLaMA 3 8B	extraction	1.000	0.0000	0.00%
Mistral 7B	screening	1.000	0.0000	0.00%
Mistral 7B	extraction	1.000	0.0000	0.00%
Gemma 2 9B	screening	1.000	0.0000	0.00%
Gemma 2 9B	extraction	1.000	0.0000	0.00%
Claude Sonnet 4.5	screening	0.974	0.0096	2.60%
Claude Sonnet 4.5	extraction	0.050	**0.8180**	**95.00%**
Gemini 2.5 Pro	screening	0.936	0.0198	6.40%
Gemini 2.5 Pro	extraction	0.200	0.4927	80.00%
GPT-4.1	screening	0.970	0.0137	3.00%
GPT-4.1	extraction	0.150	0.4858	85.00%

16 BERTScore F1: Raw vs. Rescaled (Random-Pair Baseline)

BERTScore F1 with `roberta-large` is known to saturate near 1.0 even for thematically unrelated sentence pairs because the underlying contextual embeddings cluster all natural-English text in a small subspace. The package's authors recommend rescaling F1 against a per-language random-pair baseline (`rescale_with_baseline=True`, `lang="en"`) to make absolute magnitudes interpretable. We compute both raw and rescaled F1 over the same $\binom{10}{2}$=45 run-pairs per article (4,500 pairs per model).

The rescaling substantially separates the API models: raw F1 places all three within 0.003 of each other (saturation), whereas rescaled F1 reveals that Gemini's minimum pair similarity drops

Table 16: Token-bigram Jaccard similarity on screening rationale (free-form text) vs. metadata (categorical concatenation). Computed across all $\binom{10}{2}$=45 run-pairs per item, averaged over 500 abstracts. Demonstrates that BERTScore $F_1 \geq 0.997$ on metadata is a saturation artifact: in non-saturating text (rationale), Claude exhibits only 0.378 mean similarity.

Model	**Metadata Jaccard**	**Rationale Jaccard**	**Gap**
LLaMA 3 8B	1.000	1.000	0.000
Mistral 7B	1.000	1.000	0.000
Gemma 2 9B	1.000	1.000	0.000
Claude Sonnet 4.5	0.978	**0.378**	0.600
Gemini 2.5 Pro	0.978	0.623	0.355
GPT-4.1	0.984	0.725	0.259

Table 17: BERTScore F1 (raw vs. rescaled-with-baseline) per deployment stack, computed on five-field metadata concatenations across all 4,500 run-pairs. Rescaled F1 subtracts the random-pair baseline, exposing the genuine semantic agreement signal that raw F1 saturates. Negative rescaled values (e.g., Gemini's minimum −0.458) indicate pairs less similar than random English text. Means/std/min/max are over the 4,500 pairs.

Deployment Stack	**Raw F1**			**Rescaled F1**		
	Mean	Min	≥ 0.95	Mean	Min	≥ 0.95
LLaMA 3 8B / Ollama / M4	1.000	1.000	100.0%	1.000	1.000	100.0%
Mistral 7B / Ollama / M4	1.000	1.000	100.0%	1.000	1.000	100.0%
Gemma 2 9B / Ollama / M4	1.000	1.000	100.0%	1.000	1.000	100.0%
Claude Sonnet 4.5 / Anthropic	0.997	0.931	99.7%	0.983	0.594	88.7%
Gemini 2.5 Pro / Google	0.997	0.754	98.5%	0.979	−0.458	93.4%
GPT-4.1 / OpenAI	1.000	0.956	100.0%	0.998	0.740	98.9%

below the random-English baseline (−0.458), confirming that some Gemini extraction-text pairs are genuinely semantically dissimilar—not merely subject to numeric-precision noise. This supports the main-text observation (Finding 3 of Section 4.5) that "Gemini shows the lowest minimum BERTScore F_1=0.754 ... some articles receive genuinely divergent interpretations across runs."

17 Multiple Comparison Correction for Field-Level Contrasts

The original two-proportion z-test was replaced with McNemar's test because the contrasts compare *paired* per-item consistency outcomes on the same 100 articles (or 500 abstracts) across cloud APIs. We compute, for each model M and field F, the article-level binary indicator $b_i^{M,F}=1$ if all 10 runs of M produced an identical value for F on article i, 0 otherwise; the paired contrasts are then evaluated by McNemar's test on the 2×2 discordance table of $(b_i^{M_1,F}, b_i^{M_2,F})$, using the exact mid-$p$ binomial when discordants $b+c \leq 25$ and the chi-square with continuity correction otherwise (`statsmodels.stats.contingency_tables.mcnemar`). Holm-Bonferroni (FWER) and Benjamini-Hochberg (FDR) corrections are applied to the 21 paired contrasts (5 metadata fields plus overall extraction and screening EMR; 3 pairwise comparisons each). Of 21 contrasts: 7 are significant at raw $p<0.05$, 4 survive Holm correction, and 5 survive BH-FDR.

Table 18: Paired McNemar contrasts of cloud-API consistency surviving Holm-Bonferroni or BH-FDR correction at α=0.05. "b/c" are the discordant cell counts (b=M_1 match, M_2 mismatch; c=M_1 mismatch, M_2 match).

Field	**Models**	**Δ EMR**	b/c	**Raw** p	**Holm** p	**BH-FDR** p
study_location	Claude vs. GPT-4.1	−0.220	0/22	<0.0001	**<0.0001**	**<0.0001**
study_location	Gemini vs. GPT-4.1	−0.104	0/10	0.0020	**0.0391**	**0.0136**
extraction_overall_EMR	Claude vs. Gemini	−0.150	4/19	0.0026	**0.0468**	**0.0136**
screening_overall_EMR	Claude vs. Gemini	+0.038	27/8	0.0023	**0.0446**	**0.0136**
study_location	Claude vs. Gemini	−0.115	2/13	0.0074	0.1255	**0.0310**

The remaining 16 contrasts (population, sample_size, study_design, study_period; some screening/extraction overall pairs) are not statistically robust after FWER correction; the original "distinct instability profiles" claim (main text Finding 3) is therefore tightened to: *differences are robust for study_location, overall extraction EMR (Claude vs. Gemini), and overall screening EMR (Claude vs. Gemini), but not for other categorical fields.* Replacing the original two-proportion z-test with McNemar slightly increases the number of contrasts surviving Holm correction (from 3 to 4), as McNemar conditions on the discordant pairs and gains power when within-pair concordance is high.

18 Fleiss' Kappa for Inter-Run Agreement

We report Fleiss' κ for inter-run agreement on screening decisions and extraction outputs, treating each of the 10 LLM runs as one rater. Fleiss' κ adjusts the raw concordance for the agreement expected by chance under fixed-marginal multinomial sampling, providing a more interpretable benchmark than EMR alone (which conflates "perfectly consistent" with "trivially consistent in a degenerate-marginals regime").

Two caveats govern how the table below should be read, and we state them before the numbers.

First, the chance correction is inert for the whole-output hash column. Treating each distinct output hash as a category yields 667 categories over 100 items for Claude, 265 for Gemini and 298 for GPT-4.1. With categories that numerous and that sparsely populated, the expected agreement p_e falls to $\approx$0.003–0.005, so $\kappa=(p_o-p_e)/(1-p_e)$ collapses onto p_o and reproduces the pairwise concordance already reported in Table 4 rather than adding chance-corrected information. We keep the column because the ordering across stacks is informative, and we do not interpret it against the Landis–Koch bands, which presuppose a meaningful chance baseline. The per-field columns, where categories number in the single digits, are not affected.

Second, the intervals are bootstrap, not closed-form. The classical variance of Fleiss (1971, eq. 13) depends only on the marginal proportions and is the asymptotic variance *under H_0: κ=0*, appropriate for testing that null and not for placing an interval around an estimated κ. Applied to these data it returned upper limits above 1.000 in 18 of 24 cells and assigned different standard errors to stacks whose κ was exactly 1.000. We therefore report percentile bootstrap intervals resampling items (10,000 resamples, seed 42). Where every resample returns κ=1.000 the interval is degenerate and is shown as $[1.000, 1.000]$; the informative statement for those cells is the rule-of-three bound on the non-match rate, as used for the EMR=1.000 cells in the main text.

Table 19: Fleiss' κ (95% CI) for inter-run agreement across 10 runs per deployment stack. Screening: 3-class decisions (include/exclude/uncertain) over items present in all 10 runs. Extraction (hash): each unique output hash treated as a distinct category. Extraction (per categorical field): study_design and population. Landis & Koch (1977) reference bands: < 0.0 poor, 0.0–0.2 slight, 0.21–0.4 fair, 0.41–0.6 moderate, 0.61–0.8 substantial, 0.81–1.0 almost perfect; these bands are not applied to the hash column, where the chance baseline is inert (see text). Intervals are percentile bootstrap over items (10,000 resamples, seed 42). †every bootstrap resample returned κ=1.000; the interval is degenerate and the rule-of-three bound on the non-match rate is the informative quantity.

Deployment Stack	**Screening (3-class)**	**Extraction (hash)**	**study_design**	**popula**
LLaMA 3 8B / Ollama / M4	1.000†	1.000†	1.000†	1.000
Mistral 7B / Ollama / M4	1.000†	1.000†	1.000†	1.000
Gemma 2 9B / Ollama / M4	1.000†	1.000†	1.000†	1.000
Claude Sonnet 4.5 / Anthropic	0.981 [0.970, 0.991]	**0.180 [0.133, 0.226]**	0.985 [0.952, 1.000]	0.974 [0.944
Gemini 2.5 Pro / Google	0.975 [0.961, 0.987]	**0.505 [0.445, 0.559]**	1.000†	0.979 [0.956
GPT-4.1 / OpenAI	0.974 [0.960, 0.986]	**0.511 [0.453, 0.564]**	0.990 [0.970, 1.000]	0.997 [0.989

For screening decisions, all three local stacks attain the trivial upper bound κ=1.000 and the three cloud APIs fall in the "almost perfect" band ($\kappa\in[0.974, 0.981]$). We deliberately avoid the shorthand that the stacks therefore "reach the Cochrane guideline": inter-run Fleiss' κ measures one stack reproducing itself across repetitions, whereas the Cochrane $\kappa\geq$0.80 guideline concerns agreement between two independent human reviewers. On this corpus our own two human raters reached κ=0.529 (Section 4.2), and Mistral-7B attains perfect self-consistency at a specificity of 0.240, which together show that the two quantities cannot be substituted for one another. For *whole-output* extraction hashes, the coefficient falls to 0.180 (Claude), 0.505 (Gemini) and 0.511 (GPT-4.1). Because the chance baseline is inert at this granularity, these figures should be read as the observed whole-output concordance rather than as chance-corrected agreement, and we do not attach Landis–Koch labels or compare them to the Cochrane $\kappa\geq$0.80 threshold. The ordering is nonetheless the substantive point, and it matches the EMR ranking: Claude's 10 runs reproduce a whole extraction output far less often than either other cloud stack, and none of the three approaches the local stacks. Per-field κ on categorical extraction fields (study_design, population) remains in the "almost perfect" band for all stacks, isolating the source of low whole-output κ to the variable-length `estimates` array (already documented in Section 4.4 of the main text).

19 Meta-Analytic Propagation: Full Analysis

To illustrate how extraction non-determinism propagates into downstream analysis, we performed a fixed-effect inverse-variance meta-analysis on the extracted risk ratios (RR/OR/HR) from each run independently (Table 20).

Table 20: Meta-analytic inputs across 10 runs. "k" is the number of usable effect estimates per run; "k_{sig}" is the number of those estimates whose confidence interval excludes the null. Δk and Δk_{sig} report the range across runs. The "Varying" column reports the proportion of articles for which the number of extracted estimates changed across runs.

Deployment Stack	**k range**	**Δk**	**k_{sig} range**	**Δk_{sig}**	**Varying**
LLaMA 3 8B / Ollama / M4	246–246	0	188–188	0	0/100 (0%)
Mistral 7B / Ollama / M4	229–229	0	186–186	0	0/100 (0%)
Gemma 2 9B / Ollama / M4	203–203	0	157–157	0	0/100 (0%)
Claude Sonnet 4.5 / Anthropic	186–209	23	152–175	23	21/100 (21%)
Gemini 2.5 Pro / Google	154–171	17	133–150	17	20/100 (20%)
GPT-4.1 / OpenAI	176–180	4	150–154	4	5/100 (5%)

Three findings emerge. **First**, all three local models produced identical estimate sets across all runs: LLaMA extracted 246 estimates, Mistral 229, and Gemma 203 per run, with zero variation—confirming that deterministic inference extends to the structural composition of extractions, not just field-level content.

Second, the three API models showed varying degrees of structural instability. Claude's per-run estimate count varied from 186 to 209 (a 12.4% swing, 21% of articles affected), Gemini's from 154 to 171 (11.0%, 20% of articles), and GPT-4.1's from 176 to 180 (2.3%, 5% of articles). GPT-4.1 was the most structurally stable API model, with only 5 articles producing a different number of estimates across runs—compared to 20–21 for Claude and Gemini.

Third, the variation in estimate counts directly mirrors the variation in significant estimates. For Claude, the number of significant estimates ranged from 152 to 175 (a swing of 23), meaning that **up to 23 study-level effect estimates—each representing a distinct epidemiological finding—appear or disappear from the evidence base depending solely on which run of the LLM was used**. GPT-4.1's significant estimates ranged from 150 to 154 (a swing of 4), while Gemini's ranged from 133 to 150 (a swing of 17).

While the overall pooled effect in our corpus remained stable across runs for all models (because the aggregate signal is strong and k is large), the instability of the underlying evidence composition is concerning. In domains with smaller literatures or marginal effects, even GPT-4.1's relatively modest variation (Δk=4) could plausibly shift pooled conclusions, while Claude's and Gemini's larger swings pose a more direct threat to meta-analytic reproducibility.

Random-effects re-analysis and small-literature simulation. We re-ran the propagation analysis using DerSimonian–Laird random-effects pooling on log(RR), one pooled estimate per model×run combination over the INCLUDE set. The number of articles actually entering each pool is smaller than the 100 INCLUDE articles and varies by stack, because an article contributes only when the run returned an effect estimate and a usable interval: k ranges from 39 (Gemini) to 63 (LLaMA) across the 60 combinations, and this variation in k is itself a consequence of extraction non-determinism rather than a property of the corpus. Across all 60 combinations, the pooled RR ranged from 1.012 to 1.038 (mean 1.026); the within-model range across runs was small in absolute terms (Claude 0.0057, Gemini 0.0058, GPT-4.1 0.0059) and remained within the bootstrap CI of any single run. Heterogeneity was high and stable throughout: I^2 ranged from 90.97% to 94.71% and τ^2 from 1.4×10^{-5} to 2.9×10^{-4} across the 60 combinations, so the random-effects specification is the appropriate one and the pooled estimate should be read as an average effect across a heterogeneous set rather than as a common effect. **No run-pair within any model produced a 95% random-effects CI that crossed the null** in the full-corpus pooling, confirming that, in a well-powered, large-k meta-analysis with a strong aggregate signal, LLM non-determinism does not flip the meta-analytic conclusion.

This robustness, however, depends critically on k. We subsampled the corpus to plausible small-literature scenarios ($k \in \{10, 15, 20, 30\}$, N=200 random subsamples per k, seed=42) and computed random-effects pooled estimates for each of the 10 LLM runs. For k=10 subsamples, **0.5% of Claude and Gemini subsamples** (1 of 200 each) produced *unstable null-crossing* under the DerSimonian-Laird Wald-z formulation: the pooled 95% CI crossed RR=1 in some runs but not others, meaning that the meta-analytic conclusion (significantly above null vs. not significant) **changed depending on which LLM run generated the inputs**. For $k\geq15$, no subsample exhibited unstable null-crossing across runs in any model under DL. This demonstrates a concrete, plausible scenario where LLM non-determinism affects the downstream conclusion: emerging literatures with $k\sim10$ articles (small reviews of new exposures, rare outcomes, or recent topics) are susceptible to run-dependent reversals, even when the underlying signal is detectable.

Because DerSimonian-Laird is known to be liberal in small-k settings, we re-ran the simulation with the Hartung-Knapp-Sidik-Jonkman (HKSJ) variance correction (Cochrane Handbook 6.5+ recommendation for $k<10$). Under HKSJ at k=10, the unstable-null-crossing rate rises to **31.5% (Claude), 48.5% (Gemini), and 20.0% (GPT-4.1)**, with non-trivial rates persisting up to k=20 for Claude and Gemini (16.5% and 10.5%, respectively) before vanishing at k=30. The qualitative conclusion is robust to the variance estimator (run-dependent reversals affect only small literatures); the magnitude is substantially amplified under the more conservative HKSJ formulation appropriate for $k<10$. Full DL-vs.-HKSJ comparison appears in Supplementary Section 20 and Table 22.

20 Hartung-Knapp-Sidik-Jonkman Random-Effects Sensitivity

we re-ran the per-run random-effects meta-analysis (full 100-article corpus, one pooled estimate per model×run combination) using the Hartung-Knapp-Sidik-Jonkman (HKSJ) variance correction alongside the original DerSimonian-Laird (DL) Wald-z formulation. Both estimators share the same τ^2 (method-of-moments); HKSJ inflates the standard error to $\sqrt{q^*/\sum w_i^*}$ and uses $t_{0.975,k-1}$ as the critical value. Cochrane Handbook 6.5+ recommends HKSJ over DL for $k<10$ because DL is known to be liberal (anti-conservative).

On the full 100-article corpus, HKSJ widens CIs by $2.2-2.7\times$ but every per-run CI continues to exclude RR=1 under both estimators. Thus the main-text conclusion that "no run-pair within any model produced a 95% random-effects CI crossing the null in the full-corpus pooling" is *robust* to the choice of DL or HKSJ. The HKSJ correction matters substantially in the small-literature simulation (Section 21), where the t_{k-1} critical value and the multiplicative q^* inflation become dominant.

Table 21: Per-run random-effects pooled RR (mean across 10 runs) and 95% CIs under DL (Wald-z) vs. HKSJ (t_{k-1}), with HKSJ-to-DL width ratio. "% runs cross null" = fraction of the 10 LLM runs whose pooled 95% CI on the full 100-article corpus includes RR=1.

Deployment Stack	Mean RE RR	Mean DL CI	Mean HKSJ CI	HKSJ/DL width	% runs c
LLaMA 3 8B	1.0377	$[1.031, 1.045]$	$[1.019, 1.057]$	2.65×	0% (DL) /
Mistral 7B	1.0317	$[1.025, 1.038]$	$[1.016, 1.048]$	2.51×	0% (DL) /
Gemma 2 9B	1.0253	$[1.020, 1.031]$	$[1.012, 1.039]$	2.51×	0% (DL) /
Claude Sonnet 4.5	1.0148	$[1.013, 1.017]$	$[1.009, 1.021]$	2.70×	0% (DL) /
Gemini 2.5 Pro	1.0213	$[1.018, 1.025]$	$[1.012, 1.031]$	2.54×	0% (DL) /
GPT-4.1	1.0236	$[1.019, 1.028]$	$[1.013, 1.034]$	2.22×	0% (DL) /

Table 22: Random-effects pooled RR variation across 200 random subsamples per k, seed=42, computed independently for each of 10 LLM runs. "Unstable null-crossing" = subsamples where the pooled 95% CI crosses RR=1 in some runs but not others (i.e., the meta-analytic conclusion changes depending on the LLM run). Both DerSimonian-Laird (DL, Wald-z) and Hartung-Knapp-Sidik-Jonkman (HKSJ, t_{k-1}) variances are reported; HKSJ is recommended by Cochrane Handbook 6.5+ for k<10.

Model	k	Mean range	P95 range	% unstable (DL)	% unstable (HKSJ)
Claude Sonnet 4.5	10	0.0082	0.0292	**0.50%**	**31.50%**
Claude Sonnet 4.5	15	0.0081	0.0237	0.00%	**16.50%**
Claude Sonnet 4.5	20	0.0078	0.0197	0.00%	**16.50%**
Claude Sonnet 4.5	30	0.0073	0.0150	0.00%	0.00%
Gemini 2.5 Pro	10	0.0167	0.0471	**0.50%**	**48.50%**
Gemini 2.5 Pro	15	0.0131	0.0350	0.00%	**28.50%**
Gemini 2.5 Pro	20	0.0110	0.0227	0.00%	**10.50%**
Gemini 2.5 Pro	30	0.0089	0.0151	0.00%	0.00%
GPT-4.1	10	0.0063	0.0263	0.00%	**20.00%**
GPT-4.1	15	0.0054	0.0150	0.00%	5.00%
GPT-4.1	20	0.0065	0.0149	0.00%	0.50%
GPT-4.1	30	0.0060	0.0114	0.00%	0.00%

21 Small-Literature Meta-Analytic Simulation

HKSJ sensitivity. we re-ran the small-literature simulation with the Hartung-Knapp-Sidik-Jonkman (HKSJ) variance correction alongside the original DL/Wald-z formulation. HKSJ uses the same method-of-moments τ^2 estimator as DL but inflates the standard error to $\sqrt{q^*/\sum w_i^*}$, where $q^*{=}\frac{1}{k-1}\sum w_i^*(y_i{-}\hat{\theta}_{\text{RE}})^2$, and replaces the $z_{0.975}$ Wald critical value with $t_{0.975,\,k-1}$. Because t-percentiles inflate as $k{\to}2$, and because the HKSJ variance-inflation factor satisfies $q^*{\geq}1$ whenever the observed dispersion exceeds what the fixed weights predict, HKSJ produces substantially wider CIs in small literatures. The unstable-null-crossing rate under HKSJ is markedly higher: at $k{=}10$, **31.5**% of Claude subsamples, **48.5**% of Gemini subsamples, and **20.0**% of GPT-4.1 subsamples produce run-dependent reversals of the meta-analytic conclusion (versus 0.5%, 0.5%, and 0.0% under DL). The qualitative conclusion of the main text—*LLM non-determinism propagates into the downstream conclusion only in small literatures*—is robust to the choice of variance estimator; the magnitude of the effect is substantially amplified under HKSJ, which Cochrane Handbook 6.5+ recommends precisely for $k{<}10$.

22 Seed-Effect Empirical Test

The journal's guidance for GenAI submissions requires authors to document specific random-seed values and, for commercial models, API versions and access dates; it stops short of establishing whether a seed parameter delivers determinism once documented.

Table 23: Reproducibility by seed availability and deployment paradigm. Cloud stacks: Claude Sonnet 4.5 (exposes no seed parameter, and—see the configuration table in the main text—ran at the provider's default temperature rather than at 0), Gemini 2.5 Pro and GPT-4.1 (seed=42, temperature=0 verified as transmitted). Local stacks all use seed=42 at temperature=0. The rows are descriptive: the "no seed" row differs from the "seed=42" row in provider, infrastructure, API layer and temperature simultaneously, so their difference is not an estimate of a seed effect.

Group	Stage	Mean EMR	Mean pairwise dis.
Cloud, seed=42	screening	0.953	0.017
Cloud, no seed	screening	0.974	0.010
Local, seed=42	screening	1.000	0.000
Cloud, seed=42	extraction	0.175	0.489
Cloud, no seed	extraction	0.050	0.818
Local, seed=42	extraction	1.000	0.000

The comparison that this table supports is the deployment one: local stacks at seed=42 and temperature=0 reach EMR=1.000 for extraction, while cloud stacks holding the same nominal settings reach 0.175 on average—a difference of +0.825 EMR between paradigms whose declared configuration is identical.

The comparison it does *not* support is a seed effect. The +0.125 EMR separating the seeded cloud row from the unseeded one was previously reported as such, and we withdraw that reading: those rows differ in provider, infrastructure, API layer and transmitted temperature at the same time, which is precisely the confound our unit-of-analysis framing exists to prevent. Isolating the seed requires a within-stack ablation—one provider, one infrastructure, run with and without the seed—which we did not perform.

The conclusion that survives is narrower and rests only on the seeded stacks: Gemini and GPT-4.1 transmitted seed=42 and temperature=0 on every call and still returned extraction EMR of 0.200 and 0.150. Where a seed parameter exists, is accepted, and is verifiably sent, it does not make an API-served extraction pipeline reproducible.

23 Pre-Registration Disclosure

This study is partly pre-registered, and the boundary is stated precisely. The original experiments (main text §4.1–§4.6) are exploratory: 10 repetitions, whole-output hash EMR, four semantic-tier definitions, and fixed-effect meta-analytic pooling were chosen during analysis design but without an OSF or AsPredicted submission. The dual-human labeling validation is the exception and *is* formally pre-registered as a frozen OSF registration dated 2026-05-12 (`https://doi.org/10.17605/OSF.IO/FGN3E`), two months before either label set was returned; see Section 4.2. The remaining post-revision analyses (§4.7 fixed-slot, §4.8 desconfound, §5.5 cost-benefit, silver-external) were designed and committed to the public Git repository *before* result computation; we do not present Git commits as equivalent to pre-registration.

Key commit timestamps (publicly auditable at `https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility`):

- **2026-02-11** (`75292f2`): project initialization, prompts committed
- **2026-02-13 to 2026-03-02**: original experiment raw outputs committed
- **2026-03-02** (`1b90c1b`): `analysis/reproducibility_results.json` committed (results frozen)
- **2026-03-11** (`646c6a3`): first complete manuscript draft
- **2026-03-20** (`7935445`): submission-hardening revision
- **2026-04-23 to 2026-04-28**: post-revision blindage analyses committed

We acknowledge the absence of formal pre-registration for the exploratory core as a limitation and recommend that future LLM-in-SR studies pre-register analyses on OSF before data collection.

24 Corpus Composition

Table 24: Corpus composition for the $PM_{2.5}$ and respiratory health abstract screening task.

Category	**Count**	**Purpose**
Clearly include	100	Positive controls (meet all inclusion criteria)
Clearly exclude	100	Negative controls (clear exclusion reasons)
Ambiguous	300	Primary test set (borderline cases)
Total	**500**	

Table 25: Mapping of candidate non-determinism mechanisms onto the deployment stacks evaluated in this study. ✓ = documented in public materials (e.g., provider engineering blog, vendor docs, or directly observable from API behaviour); *likely* = strongly consistent with public information about the stack but not directly attested per request; *possible* = compatible with public information but unconfirmed; — = precluded by the stack's design (e.g., single-replica local inference cannot use multi-GPU all-reduce). Attribution for the three API-served stacks is inferential; for the three local Ollama/M4 stacks, the absence of every listed mechanism follows directly from the single-device, single-execution-thread serving configuration. The two Together-AI / DeepInfra cloud columns refer to the same-weights desconfound stack (Section **??**).

Mechanism	**Local (Ollama/M4)**	**Cloud (DeepInfra/L3-8B)**	**API-Claude**	**API-Gemini**	**API-GPT-4.1**
1. Floating-point non-associativity (BF16/FP16 mixed-precision accumulation)	—	✓	likely	likely	likely
2. Tensor parallelism / multi-GPU all-reduce reduction order	—	likely	likely	likely	likely
3. FlashAttention / fused-kernel non-determinism	—	likely	likely	likely	likely
4. Dynamic continuous batching (batch composition varies between calls)	—	likely	✓	✓	✓
5. Speculative decoding / draft-model fast-paths	—	possible	likely	likely	likely
6. Prefix-caching state shared across requests	—	possible	possible	possible	possible

25 Non-Determinism Mechanisms by Stack

26 Computational Resources

27 Field-Level Extraction EMR

28 Semantic Equivalence Under Relaxed Criteria

Table 26: Computational resources per deployment stack (both stages combined, 10 runs each). Latency is the mean per-call inference duration. Cost includes API pricing and estimated electricity for local stacks (Apple M4, 15W, $0.10/kWh).

Deployment Stack	**Type**	**Inference (h)**	**Tokens (M)**	**Latency (s)**	**Cost ($)**
LLaMA 3 8B / Ollama / M4	Local	52.1	8.7	32.5	0.08
Mistral 7B / Ollama / M4	Local	76.4	10.6	46.1	0.11
Gemma 2 9B / Ollama / M4	Local	45.6	9.0	27.7	0.07
Claude Sonnet 4.5 / Anthropic	API	7.9	9.9	4.7	46.11
Gemini 2.5 Pro / Google	API	20.3	9.0	12.2	0.00*
GPT-4.1 / OpenAI	API	5.0	8.6	3.0	23.29
Total		**207.2**	**55.8**	—	**69.66**

*Gemini was used under the free tier (rate-limited).

Table 27: Field-level EMR for extraction outputs across deployment stacks. The first three columns are local Ollama/M4 stacks; the last three are API-served (Anthropic, Google, OpenAI).

Field	**LLaMA**	**Mistral**	**Gemma**	**Claude**	**Gemini**	**GPT-4.1**
Study design	1.000	1.000	1.000	0.990	1.000	0.990
Study period	1.000	1.000	1.000	0.990	1.000	0.990
Sample size	1.000	1.000	1.000	0.970	1.000	0.980
Population	1.000	1.000	1.000	0.960	0.948	0.990
Study location	1.000	1.000	1.000	0.780	0.896	1.000
Est. stability	**1.000**	**1.000**	**1.000**	**0.880**	**0.735**	**0.922**

Table 28: Semantic equivalence under progressively relaxed criteria ($n = 100$ articles, 10 runs). Columns 2–4 report whole-output EMR over the five metadata fields plus estimate count stability. BERTScore F1 (`roberta-large`) reports the mean $\pm$ SD across all $\binom{10}{2}$=45 unique run pairs per article (4,500 pairs per model). Raw F1 is reported here; rescaled F1 (against the package's random-pair English baseline) is shown in Supp. Table **??**. Hash-based EMR (from Table **??**) additionally requires identical numeric estimates. The gap between high BERTScore ($\geq$0.997) and low hash-based EMR ($\leq$0.200) demonstrates that even near-identical metadata co-occurs with divergent numeric extractions.

Deployment Stack	**Exact**	**Norm.**	**Fuzzy**	**BERTScore F1**†	**Hash***
LLaMA 3 8B / Ollama / M4	1.000	1.000	1.000	1.000 ± 0.000	1.000
Mistral 7B / Ollama / M4	1.000	1.000	1.000	1.000 ± 0.000	1.000
Gemma 2 9B / Ollama / M4	1.000	1.000	1.000	1.000 ± 0.000	1.000
Claude Sonnet 4.5 / Anthropic	0.640	0.680	0.690	0.997 ± 0.009	0.050
Gemini 2.5 Pro / Google	0.620	0.620	0.630	0.997 ± 0.022	0.200
GPT-4.1 / OpenAI	0.880	0.880	0.880	1.000 ± 0.004	0.150

*From Table **??**; includes all fields and numeric estimates.

†Raw F1 reported; rescaled F1 (against random-pair baseline) shown in Supp. Table **??**.